

IMPUTATION METHODS FOR ESTIMATING REGRESSION PARAMETERS UNDER A MONOTONE MISSING COVARIATE PATTERN: A COMPARATIVE ANALYSIS

A. Satty

School of Mathematics, Statistics and Computer Science, University of
KwaZulu-Natal, Private Bag X01 Scottsville 3209, Pietermaritzburg, South
Africa

and

H. Mwambi

School of Mathematics, Statistics and Computer Science, University of
KwaZulu-Natal, Private Bag X01 Scottsville 3209, Pietermaritzburg, South
Africa

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Summary: The paper deals with the problem of missing covariate values in a regression model. An attractive approach to avoid this problem is to impute the missing covariate values rather than delete cases with missing covariates. The paper is devoted to a comparison of different imputation techniques or methods. The type of missing data pattern in the set of independent variables is the monotone data pattern. We assume that the missing data are missing at random (MAR). Imputation of missing values was achieved using a multivariate normal model. The main objective of this paper is to study how some imputation approaches compare when the missing data pattern is monotone. The techniques that were considered include the Last Observation Carried Forward (LOCF), Propensity Score (PS), Markov Chain Monte Carlo (MCMC) and Regression. In order to compare the performance of the proposed methods, we used originally complete data sets (no data are missing), and then we intentionally created missing values to achieve the intended goal. Missingness was imposed on covariate variables. The performance of these

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four methods is compared on three criteria: bias, efficiency and coverage. Data from a Diabetes study is used to illustrate the considered approaches.

1. Introduction

The occurrence of missing data in scientific investigations is quite common. Data analysis is seldom performed in the absence of missing data, which brings to question efficiency of estimates, validity of analysis and bias due to differences between observed and unobserved data (Little (1992)). To this end, considerable research has recently focused on finding appropriate procedures for handling missing data, see, for example, Zhang (2003), Chen (2004), Horton and Kleinman (2007), Peng and Zhu (2008) and Chen (2002). One approach for handling incomplete data problems that addresses these concerns is imputation, which was proposed by Rubin (1978) and described in detail by Little and Rubin (1987), Little and Rubin (2002), and Schafer (1997). Schafer (1999) provides extensive bibliography on missing data imputation approaches.

Regression analysis users are often faced with the problem of dealing with a number of missing values in one or more explanatory variables. The easy solution is often to discard the cases of the missing value variables in the model and to confine attention to complete case analysis. However, this solution, although simple and straightforward, usually leads to adverse statistical consequences and is surely not an efficient strategy. An alternative more efficient approach is to impute new values that replace the missing ones before carrying out the regression analysis. See, e.g., Little (1976) for basic considerations and Little (1992) for detailed discussion of regression analysis with imputed values for missing data. The reasons for missing data are

commonly called missing data mechanisms. According to Little and Rubin (1987) and Rubin (1976), the missing data mechanisms can be classified as: missing completely at random (MCAR), missing at random (MAR), and missing not at random (MNAR). The mechanisms are discussed briefly in later sections.

This paper focuses on comparison of several imputation techniques applied to an incomplete data set under MAR. We focus on the problem of missing covariate values in a regression model. Imputation of missing values was achieved using a multivariate normal model. The imputation techniques that will be compared include the controversial Last Observation Carried Forward (LOCF) method and the more sound multiple imputation methods implemented in the SAS software PROC MI (Markov Chain Monte Carlo (MCMC), Regression and Propensity Score (PS)). The main objective of this paper is to study how some imputation approaches compare when the missing data pattern is monotone. In the application, a monotone missing data pattern was created from complete data. The results from complete data were presented and used as a references against which these four techniques were contrasted. Data from a Diabetes study is used to illustrate the considered approaches. MATLAB and SAS programs were used to create the missing data and to implement imputation techniques, respectively. A detailed discussion on the computer programmes used for implementing missing value imputations can be found in Horton and Lipsitz (2001) with particular focus on the REG, MI and MIANALYZE procedures in SAS.

The rest of this paper is organized as follow: In Section 2 we consider the missing data patterns and missing data mechanisms. In Section 3, we review the methods of imputation to be used: LOCF, Regression, Propensity score and MCMC. For each method a brief literature review is given, as well

as the description of the multiple imputation for continuous data. In Section 4, we present an application including a description of the full data set used in the analysis. The design of the comparative study used in the analysis is discussed in details. The results from the application of the different methods are presented in Section 5. Finally, in Section 6, we conclude by a discussion of the results.

2. Missing data patterns and classifications

2.1 Missing data patterns

An important concept with missing data, specifically where there are multiple variables with missing values, relates to the pattern of missing data. There is a need to identify the pattern of the missing data because some methods apply to missing data in general, while others are restricted to a specific kind of pattern. In this paper, we focus on the monotone missing pattern, as defined by Little and Rubin (1987), Rubin (1974) and Anderson (1957), meaning that a data matrix is said to have a monotone missing pattern if, whenever an element y_{ij} is missing, the element y_{ik} are also missing for all $k > j$. The notation y_{ij} means the i th observation for variable j , where $i = 1, \dots, n$ and $j = 1, \dots, p$.

For example, given variables Y_1 , Y_2 , and Y_3 , a data set is said to have a monotone missing patterns if the missing information in the variables is ordered in some specific way. For illustration, Table (1) represents variables Y_1 , Y_2 , and Y_3 with observations 1, 2,...,8 assuming $n=8$, where an X denotes available data and a dot (.) denotes a missing data value. When an element in Y_2 is missing, so is the corresponding element in Y_3 , leading to a monotone missing data pattern (Rubin (1974)). In monotone missing pattern the ordering of variables is important. Simpler imputation methods can be used if the pattern is monotone, though a monotone pattern is uncommon in most

complex investigations. It may be possible, however to create a monotone missingness pattern that separates out a small number of observations that are non-monotone.

Table 1. Monotone and non-monotone pattern of missingness

Obs	Monotone			Non-monotone		
	Y_1	Y_2	Y_3	Y_1	Y_2	Y_3
1	X	X	X	X	.	X
2	X	.	.	X	.	.
3	X	.	.	X	.	.
4	X	.	.	X	.	.
5	X	X	.	X	X	.
6	X	X	.	X	X	.
7	X	X	.	X	X	.
8	X	X	X	X	X	X

In the non-monotone example in Table (1), all but the first observation can be rearranged into a monotone pattern (i.e., 81% of the dataset). This type of hybrid pattern is exploited by a number of computer packages. It is important to note that in many realistic settings, datasets may have missing outcomes as well as missing predictors (Chen (2004)).

2.2 Missing data classifications

As mentioned above, in this subsection we discuss three types of missing data mechanisms that are used extensively to refer to missing data in literature. These refer to mechanisms that could have generated the missing data. When a dataset has missing values, the difficulty of obtaining valid parameter estimates depends on the mechanism that causes values to be missing. A useful classification for missing value mechanisms was introduced by Little and Rubin (1987). More formally, let Y be an $n \times p$ data matrix, $Y = (y_1, y_2, \dots, y_n)^T$, where $y_i = (y_{i1}, \dots, y_{ip})^T$ is a random sample from a

p -dimensional multivariate probability distribution $P(Y \mid \theta)$ governed by parameters θ . In the following, we refer to the rows of Y as observations, given by $y_i^T (i = 1, 2, \dots, n)$, and the columns of Y as variables, denoted by $Y_j (j = 1, \dots, p)$. Moreover, we define an $n \times p$ missingness indicator matrix $R = (r_{ij})$ by letting

$$R_{ij} = \begin{cases} 1 & \text{if } Y_{ij} \text{ is missing,} \\ 0 & \text{otherwise.} \end{cases} \quad (1)$$

Defining $P(r_{ij} = 0 \mid y_{ij}) = P(y_{ij} \text{ observed} \mid y_{ij}) = p_{ij}$, then R is subject to a probability distribution $P(R \mid \psi, Y)$ governed by parameters ψ . Under this assumption, the joint probability distribution of the response variables and the missingness indicator variables can be factorized as

$$P(Y, R \mid \theta, \psi) = P(Y \mid \theta)P(R \mid \psi, Y), \quad (2)$$

where $P(Y \mid \theta)$ is the marginal distribution of the response variables, and $P(R \mid \psi, Y)$ is the conditional distribution of missingness given the response variables. Following Little and Rubin (1987) and Rubin (1987), we use the notation Y_{obs} and Y_{mis} to represent the observed portion and the missing portion of Y , i.e., $Y_{obs} = (y_{ij} \mid r_{ij} = 0)$ and $Y_{mis} = (y_{ij} \mid r_{ij} = 1)$. We also use the notation $Y_{obs,j}$ and $Y_{mis,j}$ to represent the observed portion and the missing portion of variable Y_j , and use $y_{i(obs)}$ and $y_{i(mis)}$ to represent the observed portion and the missing portion of the i th observation. In the probability model (2), there are two sets of parameters, the parameters of interest θ and the nuisance or missingness parameters ψ . The correct inferences on θ in general need to be conducted based on the joint probability model (2). Hence this inference depends on how the probability model for the missingness is defined, i.e., how the missingness depends on Y . Rubin (1976, 1987) categorized the missingness mechanism into the following three categories based on the conditional distribution $P(R \mid \psi, Y)$.

- If $P(R \mid \psi, (Y_{obs}, Y_{mis})) = P(R \mid \psi)$, the missingness is independent of the responses (observed and missing), then the missingness mechanism is defined as missing completely at random (MCAR).
- If $P(R \mid \psi, (Y_{obs}, Y_{mis})) = P(R \mid \psi, Y_{obs})$, the missingness is independent of the missing responses given the observed values. In this case the missingness mechanism is defined as missing at random (MAR).
- If $P(R \mid \psi, (Y_{obs}, Y_{mis})) \neq P(R \mid \psi, Y_{obs})$, the missingness depends on both observed and missing responses, and is defined as missing not at random (MNAR).

Note that the parameterization ψ is not expected to be the same under the three missingness categories. Of the above definitions, MCAR is the most restrictive because the missing values do not depend on the response variables, neither observed values nor missing values. In this case, the missing values for a variable are like a simple random sample of the data for that variable, so the distribution of missing values is the same as the distribution of observed values. In contrast to MCAR, MAR is a less restrictive assumption because the missing values can depend on the response variables through the observed values. In this case, the missing values for a variable are like a simple random sample of the data for that variable within subgroups defined by the observed values, and the distribution of missing values is the same as the distribution of observed values within each subgroup. The two sets of parameters θ and ψ are said to be distinct by Little and Rubin (1987), Rubin (1987) and Madow et al. (1983) if from a: 1) Frequentist perspective, the joint parameter space of (θ, ψ) is the Cartesian cross-product of the parameter spaces for θ and ψ . 2) Bayesian perspective, the joint prior distribution of (θ, ψ) can be factored into the independent marginal prior distributions for θ and ψ . This

assumption is intuitively reasonable in many real situations since knowing ψ provide no information about θ and vice versa. If θ and ψ are distinct, and if either MCAR or MAR holds, the missingness mechanism is said to be *ignorable* (Little and Rubin (1987) and Rubin (1976, 1987)), in the sense that the missingness mechanism can be ignored when making likelihood-based or Bayesian statistical inferences on the parameters of interest θ . A missingness mechanism that does not satisfy this definition is called *non-ignorable*, and the data is said to be MNAR.

3. Methods to handle missing covariates

There are a variety of imputation methods that can be used to deal with missing covariates. The subsections that follow provide a review of the methods that are used in this study.

3.1 Last observation carried forward (LOCF)

The LOCF method is a well-known, commonly used imputation technique. It retains all of the originally randomised subjects, eliminate missing data, and produces a completed data set. In this approach every missing value is replaced by the last observed value from the same subject or time series, i.e. it is a method that assumes that the outcomes would not have changed from the last observed value. Very strong and unrealistic assumptions have to be made to ensure the validity of this method. LOCF analysis appeals through its simplicity and ease of application, but there are strong grounds for not using it. Specifically, the method may introduce bias in the results, and this bias can, according to circumstance, be in either direction (Molnar et al. (2008)). The technique does not itself indicate a particular type of analysis for the resulting data. Molenberghs and Kenward (2007) noted that LOCF can be applied to both patterns of missing data (monotone and non-monotone). We refer to

Craig et al. (2003) and Siddiqui and Ali (1998) for more details, and where insightful illustrations of the issues of this method are provided in Molenberghs and Kenward (2007).

3.2 Multiple imputation for continuous data (MI)

MI was first proposed by Rubin (1978), and was originally developed as an alternative to earlier single imputation approaches by Madow et al. (1983). It has been discussed thoroughly elsewhere, see, for example, Schafer (1997), Schafer (1999), Graham and Schafer (2002), Sinharay et al. (2001) and Zhang (2003). MI refers to a procedure of replacing each missing value by a vector of $M \geq 2$ imputed values selected to reflect the uncertainty in the imputation (Rubin (1987)). The technique accounts for uncertainty in sampling from a population by introducing randomness to imputations and creating M imputed data sets, each of which is then subjected to the desired statistical analysis (e.g. regression analysis). MI incorporates information from other variables into the imputation process in order to provide more accurate values. The MI method as is used in most common implementations (e.g. PROC MI in SAS software) assumes that the method is valid under MAR. Rubin (1987) described multiple imputation as a three step process. First, sets of plausible values for missing observations are created that reflect uncertainty about the non-response model. Each of these sets of plausible values can be used to fill-in the missing values and create a completed data set. Second, each of these completed data sets can be analysed using complete-data methods. Finally, the M results are combined using methods that allow for uncertainty regarding the imputation to be taken into account.

For continuous data, multiple imputation requires an assumption that the method is valid under a multivariate normal distribution. Based on this probability model, parameter estimates are obtained using the Bayesian

posterior distribution based upon the likelihood function of the proposed model, the observed data and a prior distribution. The Markov Chain Monte Carlo (MCMC) method of data augmentation is used to obtain this posterior distribution from which the imputed values for the missing observation are drawn. This imputation process is repeated M times (e.g. 5) to create independent data sets, each of which is then subjected to the analysis of interest, such as the regression analysis as in this study. The results of the M separate analyses (e.g. parameter estimates) are then combined into a single value as

$$\bar{Q} = \frac{1}{M} \sum_{m=1}^M \hat{\theta}_m, \quad (3)$$

where $\hat{\theta}_m$ is the parameter estimate of interest from imputation $m = 1, 2, \dots, M$. The variance for these estimates is composed of two parts: the between imputation variance and within imputation variance. Between imputation variance takes the form

$$B = \frac{1}{M-1} \sum_{m=1}^M (\hat{\theta}_m - \bar{Q})^2. \quad (4)$$

The within imputation variance, $\bar{U}$, is the mean of estimated variances across the M imputations. The total variance for MI is then calculated as

$$T = \bar{U} + \left(1 + \frac{1}{M}\right) B. \quad (5)$$

In this article, we consider only the multiple imputation methods that are implemented in SAS software PROC MI. The following three methods are available in the MI procedure:

3.2.1 Markov chain monte carlo (MCMC)

MCMC method is a Monte Carlo integration method using Markov Chains (Zhang (2003)). It has been applied in many statistical situations, see, Gilks et al. (1996). Schafer (1997) applied the MCMC method for the purpose of

multiple imputations by utilizing the data augmentation algorithm developed by Tanner and Wong (1987). This method is the most popular multiple imputation method for handling missing data. This is the default method in SAS PROC MI. The technique is valid under the assumption of multivariate normality (Robins et al. (1994)), which implies that valid imputations may be generated by linear regression equations. The MCMC method is based on draws of pseudo random samples from a target probability distribution. When missing data have a non-monotone pattern, the target distribution is the joint conditional distribution of Y_{mis} and θ given Y_{obs} ,

$$P(Y_{mis}, \theta \mid Y_{obs}). \quad (6)$$

The MCMC method for imputing missing data is conducted as follows: Replace the missing data Y_{mis} by some assumed values, then θ can be simulated from the resulting complete data posterior distribution $P(\theta \mid Y_{obs}, Y_{mis})$. If $\theta^{(t)}$ is the current simulated value of θ from the complete data posterior distribution, then the next iterative sample of Y_{mis} , $Y_{mis}^{(t+1)}$, can be drawn from the conditional predictive distribution of Y_{mis} given Y_{obs} and $\theta^{(t)}$, i.e.,

$$Y_{mis}^{(t+1)} \sim P(Y_{mis} \mid Y_{obs}, \theta^{(t)}). \quad (7)$$

Conditioning on $Y_{mis}^{(t+1)}$, the next simulated value of θ can be drawn from its complete data posterior distribution,

$$\theta^{(t+1)} \sim P(\theta \mid Y_{obs}, Y_{mis}^{(t+1)}). \quad (8)$$

Repeating the random draws of Equations (7) and (8) from a starting value of $\theta^{(0)}$ yields a Markov chain $\{(\theta^t, Y_{mis}^{(t)}) : t = 1, 2, \dots\}$. The stationary distribution of this chain is the joint distribution of θ and Y_{mis} given Y_{obs} , $P(Y_{mis}, \theta \mid Y_{obs})$. Consequently, the marginal stationary distributions of the subsequence $\{\theta^{(t)} : t = 1, 2, \dots\}$ and $\{Y_{mis}^{(t)} : t = 1, 2, \dots\}$ are the observed-

data posterior distribution $P(\theta \mid Y_{obs})$ and the posterior predictive distribution $P(Y_{mis} \mid Y_{obs})$ respectively. When t is sufficiently large, $\theta^{(t)}$ can be viewed as a single simulation from the observed data posterior distribution $P(\theta \mid Y_{obs})$, and $Y_{mis}^{(t)}$ can be viewed as a single imputation from the posterior predictive distribution $P(Y_{mis} \mid Y_{obs})$. The random draw of Equation (7) is used to impute the missing data Y_{mis} , and the random draw of Equation (8) is used to simulate the unknown parameter θ . Therefore, Equations (7) and (8) are referred to as the Imputation or I-step and the Posterior or P-step respectively (Tanner and Wong (1987)). This algorithm was first used by Li (1988) who presented an argument for convergence and used it to impute the missing data Y_{mis} .

3.2.2 Regression method

A regression model is fitted for each variable with missing data, using the remaining variables as covariates (Yuan (2000)). Based on the fitted regression coefficients, a new regression model is formulated from a Bayesian predictive distribution of the parameters (regression parameter estimates and associated covariance matrix) which is then used to impute the missing values for each variable (Rubin (1987)). Let Y_j be a continuous variable with missing values, satisfying the model

$$E(Y_j) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k, \quad (9)$$

which is fitted using observations with observed values for the variable Y_j and its covariates $X_1, X_2, X_3, \dots, X_k$, where k is the remaining numbers of variables, and $0 < k < p$. According to Rubin (1987), this method assumes multivariate normality, therefore, the above model implies a Normal error model with mean 0 and variance σ_j^2 . The fitted model provides the regression parameter estimates $\hat{\beta} = (\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k)$ and the associated covariance matrix

$\hat{\sigma}_j^2 V_j$, where V_j is the j th diagonal element of the usual $(X'X)^{-1}$ matrix derived from the intercept and covariates $X_1, X_2, \dots, X_k$

Steps for generating imputed values for each variable:

1. New parameters $\beta_* = (\beta_{*0}, \beta_{*1}, \dots, \beta_{*(k)})$ and σ_{*j}^2 are drawn from the posterior predictive distribution of the parameters. That is, they are simulated from $(\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k), \sigma_j^2$ and V_j . The variance is drawn as

$$\sigma_{*j}^2 = \hat{\sigma}_j^2 (n_j - k - 1) / g, \quad (10)$$

where g is a $\chi_{n_j - k - 1}^2$ random variate and n_j is the number of non-missing observations for Y_j . The regression coefficients are drawn as

$$\beta_* = \hat{\beta} + \sigma_{*j} V'_{hj} Z, \quad (11)$$

where V_{hj} is the upper triangular matrix in the cholesky decomposition $V_j = V'_{hj} V_{hj}$ and Z is a vector of $k + 1$ independent random standard normal variates.

2. The missing values are then replaced by

$$\beta_{*0} + \beta_{*1}x_1 + \beta_{*2}x_2 + \dots + \beta_{*(k)}x_k + z_i\sigma_{*j},$$

where $x_1, x_2, \dots, x_k$ are the values of the covariates and Z_j is a simulated standard normal deviate.

3.2.3 Propensity score method (PS)

Another imputation method available for continuous variables is the propensity score method. PS method for multiple imputations was proposed by Lavori et al. (1995). Basically, this method follows a nonparametric approach in which the missing values are imputed by resampling of the observed values. The method is also valid under the assumption that the data set has a monotone missing pattern. In medical research it can be defined as the conditional probability of assignment to a particular treatment given a vector of observed covariates (Rosenbaum and Rubin (1983)). For each variable with

missing values, propensity scores are generated for all observations to estimate the probabilities that each observation is missing. For each group Bayesian imputation technique is applied.

By defining the missingness indicator variables r_{ij} as in Equation (1), the missingness can be predicted by a linear logistic regression model. The conditional probability of observing y_{ij} , given the previous history $y_{i1}, \dots, y_{i,j-1}$, can be called a propensity score s_{ij} . According to Rosenbaum and Rubin (1983) definition of the PS, let

$$s_{ij} = Pr(r_{ij} = 0 \mid y_{i1}, \dots, y_{i,j-1}). \quad (12)$$

Since the missing data have a monotone pattern, the propensity score can be modeled by

$$\log \left(\frac{s_{ij}}{1 - s_{ij}} \right) = \beta_0 + \beta_1 y_{i1} + \dots + \beta_{j-1} y_{i,j-1}, \quad (13)$$

where $\beta_0, \beta_1, \dots, \beta_{j-1}$ are the regression coefficients. After the regression coefficients in Equation (13) are estimated from the observed r_{ij} for the response variable Y_j and the complete data for the covariates $Y_1, \dots, Y_{j-1}$, each observation can be assigned an estimated propensity score,

$$\hat{s}_{ij} = \frac{\exp(\hat{\beta}_0 + \hat{\beta}_1 y_{i1} + \dots + \hat{\beta}_{j-1} y_{i,j-1})}{1 + \exp(\hat{\beta}_0 + \hat{\beta}_1 y_{i1} + \dots + \hat{\beta}_{j-1} y_{i,j-1})}, \quad (14)$$

and then all observations are stratified into q strata based on the quantiles of estimated propensity scores. Within each stratum, a 'donor pool' is created by applying the approximate Bayesian bootstrap (Rubin and Schafer (1986)). That is, a random sample is created by random draws with replacement from the observed values of Y_j within each stratum. The approximate Bayesian bootstrap method is applied in order to reflect additional uncertainty about the posterior distribution of the underlying parameters, given the observed values of Y_j within each stratum. This method is roughly equivalent to choosing the

values of the parameters θ for the conditional posterior predictive distribution $P(Y_{mis} \mid Y_{obs}, \theta)$ from the observed-data posterior distribution $P(\theta \mid Y_{obs})$. After the donor pools are created, each missing value of Y_j is then imputed by a single random draw from its donor pool. M sets of multiple imputations are obtained by creating M conditionally independent donor pools for each individual missing value and then taking a random draw from each donor pool. Note that imputing a missing value by a random draw from its stratum rather than from its donor pool would result in an improper multiple imputation in the sense that the between imputation variance would be underestimated because the uncertainty due to selecting the imputation model is not incorporated into the imputation.

4. Application

4.1 Example: Diabetes data

We consider a response variable Y that is fully observed, while all the explanatory variables $X_1, X_2, \dots, X_p$ contain missing values. Let X be the design or model matrix, of dimension $(n \times p)$, while Y represents the vector of response values of length n . The vectors $(Y_i, X_{i1}, \dots, X_{ip})$ for $i = 1, \dots, n$ are independent. Due to missing observations, the design matrix can be split into two parts, $X = (X_{obs}, X_{mis})$; X_{obs} represents the part of the design matrix X with covariates that are completely observed and X_{mis} is the subset of X with

explanatory variables which have at least one value that is not observed.

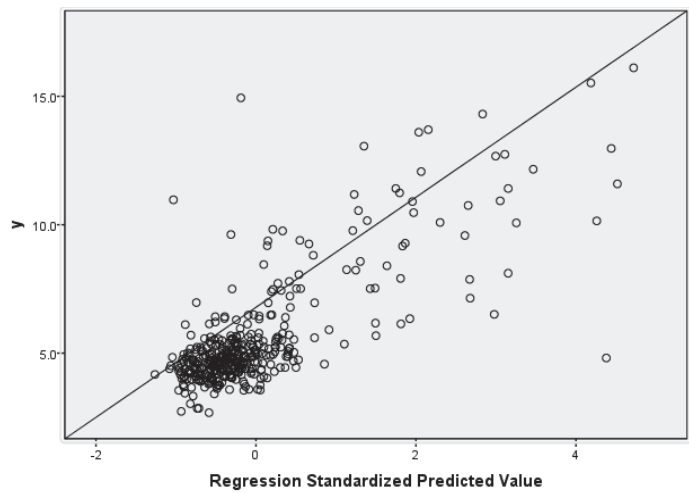


Figure 1(a). Scatterplot - complete data set.

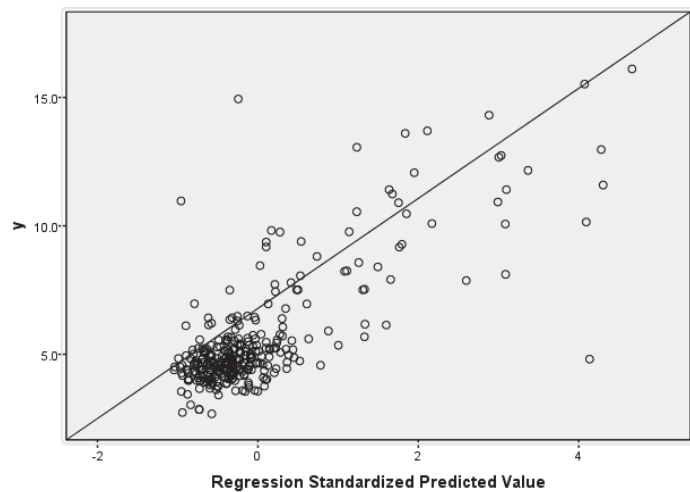


Figure 1(b). Scatterplot - 20% missing covariates.

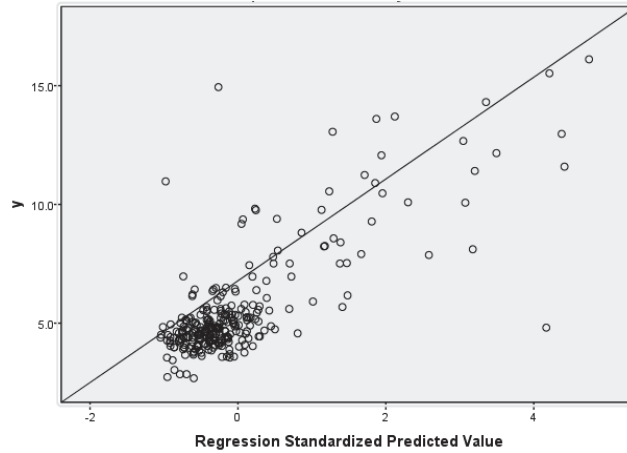


Figure 1(c). Scatterplot - 30% missing covariates.

We assume that X_{mis} follows a monotone pattern. To illustrate this application we describe a medical data set, as reported by Willems et al. (1997). The original (diabetes data set example) database consists of 19 variables on 403 subjects out of 1040 subjects who were originally interviewed in a study to understand the prevalence of obesity, diabetes and other cardiovascular risk factors in Central Virginia for African Americans. The 403 subjects were the ones who were actually screened for diabetes. At this point we have to make it clear that our present study is essentially a model comparative study rather than original case study. We compare the various imputation methods using generated missing values in a complete subset of the whole database. So, our initial strategy was to find a dataset with no missing values on certain important continuous variables and then to use it for our study. We thus ended up with a complete data set with 400 subjects. The variables which we selected are: Y - glycosolated hemoglobin, X_1 - age, X_2 - cholestrol, X_3 - high density lipoprotein, X_4 - cholestrol/HDL ratio and X_5 - stabilized glucose. We use a linear regression model for the response variable, Y , as:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 + \epsilon, \quad (15)$$

where $\epsilon \sim \text{Normal}(0, \sigma^2)$, therefore,

$$E(Y) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5, \quad (16)$$

where $E(Y)$ denotes the expectation of Y for given $X_1, \dots, X_5$. The model assumptions also imply

$$\epsilon = Y - E(Y \mid X_1, X_2, X_3, X_4, X_5) \sim \text{Normal}(0, \sigma^2), \quad (17)$$

where glycosolated hemoglobin (Y) was used as a surrogate outcome for diabetes because of its known high predictive capacity with respect to type two diabetes. In this application, all covariates ($X_1 - X_5$) were subjected to missing observations while Y was left completely observed for all cases. In order to create a monotone missing data pattern, we assumed that if a datum is missing, all subsequent data on that observation is also missing. Thus, the incomplete data set is constructed from the complete data set and the process yields a monotone missing pattern in variables $X_1 - X_5$. Figure 1 represents the relationships between the outcome Y and regression standardized predicted values of Y before and after creating missing observations.

4.2 Study design

To investigate the performance of the proposed methods, five steps were planned and executed as follows:

- We fit a linear regression model using the complete data set with the Y as the dependent variable and X_1 to X_5 as explanatory variables to obtain the regression analysis coefficients.
- From the complete dataset, we draw 1000 random samples of $N = 400$. We create a monotone missing data pattern in X_1 to X_5 . The monotone pattern missingness was created by randomly selecting values from the data matrix, then the pattern was created by discarding values that lie next to the selected

ones. Missing data for X_1 to X_5 are generated assuming (MAR) mechanism. To achieve MAR mechanism, we assume the missing data is related to the observed data, i.e., (past responses). Missing data were created in X_1 by randomly deleting 20% and 30% of observations greater than 55, resulting in monotone missingness patterns in incomplete data. A monotone missing pattern was used, which is to say that for X_1 , if a X_1 's observation is deleted, the subsequent observations in the other variables for that subject were also deleted. To generate the missing data, we use a function written in the MATLAB software package.

- We fill in the missing values in X_1 to X_5 to generate completed data sets under a multivariate normal model using imputation methods described above. For multiple imputation techniques, we used the SAS software. The MI procedure options were set so that the program used up to 1000 iterations of the EM method to find the posterior mode. We used the defaults of the SAS MI algorithm: a Jeffreys prior and initial sampler values from the EM posterior mode. Then, the method uses Markov-chain Monte Carlo (MCMC) to generate the actual imputations. PROC MI was applied to generate $M = 5$ complete data sets (conventional wisdom suggests taking $M = 5$, see, for example, Schafer (1999)). PROC REG was used to set up effect parameterizations for the variables and the BY statement is also used to allow the analysis to be repeated for each data set. The simpler LOCF technique replaced the missing data by the last available observed data, and once the data set has been completed in this way, it is analysed as if it were fully observed.
- We re-estimate the regression model using each imputed dataset separately and we record the results of interest. For the MI techniques, we combine the results of the 5 imputations using formulas that account for variation within and between the imputed datasets using Equations (3), (4) and (5).

We use PROC MIANALYZE to combine the estimates from the 5 completed datasets to generate valid statistical inferences about the regression model.

- Finally, we compare the results of imputation techniques from the fourth step with the results of the true complete data set from the first step to assess the performance of the proposed methods.

4.3 Criteria of performance

We assess the performance of the methods largely using the criteria recommended in Graham and Schafer (2002) and Peng and Zhu (2008), namely, bias, efficiency and coverage. In this article, we defined bias as the difference between the average of the 1000 coefficient estimates and the corresponding true coefficient. Thus a better approach that does on the average presents the population value with less bias. We defined the efficiency as the variability of the estimates around the true population coefficient. It was measured in this study by the average width of the 95% confidence interval. Thus a wider interval implies a less efficient technique. Coverage was defined as the percentage of 95% confidence intervals estimates across 1000 replicates. If a method is working well, the actual coverage should be close to the nominal rate (95%).

5. Results

Table (2) shows the overall performance of the imputation techniques considered in this study when the missing data rate was 20%. For all covariates the greatest bias, also the worst, is highlighted. Examining these results we find the following. When compared with the results based on the PS and LOCF methods in Table (2), the MCMC and Regression methods offered better performance than the PS and LOCF methods. Covariates (X_1 to X_5), were more accurately estimated by MCMC and Regression methods as compared to

Table 2. Bias, Efficiency and Coverage of MCMC, Regressions, PS and LOCF, under 20% missing covariates

Rate	Method	Parameter	Bias	Efficiency	coverage true level=95%
MCMC					
		X ₁	0.002	0.020	0.972
		X ₂	-0.003	0.011	0.978
		X ₃	0.000	-0.001	0.991
		X ₄	0.042	0.120	0.967
		X ₅	0.001	0.007	0.991
Regression					
		X ₁	0.002	0.021	0.973
		X ₂	-0.003	0.012	0.978
		X ₃	0.000	-0.004	0.992
		X ₄	0.061	0.120	0.961
		X ₅	0.001	0.007	0.991
PS					
		X ₁	-0.005	0.030	0.954
		X ₂	-0.003	0.012	0.968
		X ₃	0.008	-0.014	0.951
		X ₄	-0.139	0.121	0.942
		X ₅	0.004	0.013	0.966
LOCF					
		X ₁	-0.011	0.030	0.954
		X ₂	0.016	0.035	0.941
		X ₃	-0.043	-0.113	0.897
		X ₄	-0.630	0.159	0.784
		X ₅	-0.678	0.211	0.677

Note: The largest bias and efficiency for each given estimate appear in bold.

PS=propensity score; LOCF=last observation carried forward;

MCMC=markov chain monte carlo.

those of PS and LOCF. In particular, covariate X_3 was unbiased for MCMC and Regression methods. Furthermore, MCMC and Regression methods yielded the same estimates for all covariates, except for X_4 . Their respective estimates did not differ significantly from those of the true actual data.

Table 3. Bias, Efficiency and Coverage of MCMC, Regression, PS and LOCF, under 30% missing covariates

Rate	Method	Parameter	Bias	Efficiency	coverage true level=95%
MCMC					
		X_1	0.006	0.021	0.982
		X_2	-0.001	0.008	0.984
		X_3	0.001	0.003	0.977
		X_4	-0.059	0.361	0.955
		X_5	0.001	0.008	0.987
Regression					
		X_1	0.006	0.022	0.981
		X_2	-0.001	0.008	0.984
		X_3	0.000	0.003	0.994
		X_4	-0.051	0.345	0.952
		X_5	0.001	0.007	0.984
PS					
		X_1	0.013	0.033	0.950
		X_2	-0.002	0.010	0.969
		X_3	-0.005	-0.011	0.965
		X_4	0.001	0.206	0.947
		X_5	0.010	0.006	0.959
LOCF					
		X_1	-0.012	0.030	0.952
		X_2	-0.021	-0.046	0.933
		X_3	-0.046	0.126	0.877
		X_4	-0.738	0.240	0.653
		X_5	-0.556	0.216	0.755

Note: The largest bias and efficiency for each given estimate appear in bold.

PS=propensity score; LOCF=last observation carried forward;

MCMC=markov chain monte carlo.

Differences were never more than 0.003, with one exception - the estimate of X_4 , which was slightly different than that from true complete data. Both, PS and LOCF contained more biased estimates as compared to MCMC and Regression methods of imputation used in this study. Covariates which showed most bias were X_5 under the PS method and (X_1 , X_2 , X_3 and X_4) under

the LOCF method. Since a wider interval implied a less efficient, thus the widest also implies the worst, 95% is highlighted. The LOCF was less efficient most frequently. This should not be a surprise because intuitively the LOCF's weakness is that it tends to create inflated artificial information than truly expected. Between the two LOCF and PS methods, the LOCF was less efficient than the PS. The exception was with the estimate of X_1 for the PS. For X_1 to X_5 , MCMC and regression methods uniformly the best approaches in terms of efficiency. Therefore, both MCMC and Regression methods were more efficient than PS and LOCF. In terms of coverage, according to Graham and Schafer (2002), the performance of a methods can be regarded to be poor if its coverage drops below 90% and hence leads to substantially increased Type-I error rate. By this rationale, the MCMC, Regression and PS methods yielded equally acceptable performance across all covariates. The LOCF's coverage at 95% was consistently lower than 90%, except for X_1 and X_2 , thus, this coverage was indicative a seriously low level of coverage because 90% corresponds to a doubling of the nominal rate of error (0.05).

With 30% missingness, the results from the performance of the four methods are listed in Table (3). For all four methods, the MCMC and Regression methods yielded equally good performance and outperformed the PS and LOCF methods. In terms of bias condition, the benefits of MCMC and Regression methods over a PS and LOCF are clearly evident. In particular, the estimates from PS and LOCF for X_1 and (X_2 , X_3 , X_4 and X_5), respectively, contain seriously biased estimates. Both MCMC and Regression methods for X_1 , X_2 and X_5 yielded the same estimates of regression coefficients. Thus, while there appeared to have been little or no bias, they were more efficient than those obtained from the PS and LOCF. An examination of the efficiency suggested that the estimates from MCMC and Regression methods

were typically lower than those from the PS and LOCF methods, thus such estimates were more efficient than were those based on the PS and LOCF methods. Efficiency by PS and LOCF approaches appeared to be independent of the missingness rate. The worst performance on efficiency occurred with estimates of covariates X_1 and X_2 for PS, and with estimates of covariates X_3 , X_4 and X_5 for LOCF when compared with those based on MCMC and Regression methods. Consequently, both methods, being asymptotically less efficient. In comparison with results for PS, LOCF's efficiency was worst than that of PS. With respect to coverage, similar to the findings obtained under 20% missing data, the MCMC, Regression and PS method produced uniformly acceptable coverage; none was less than 90%. On the other hand, coverage rates obtained by the LOCF method in most cases were unsatisfactory with two exceptions - coverage rates for X_1 and X_2 . It is appeared that its low coverage rates can also be attributed to its large biases.

6. Discussions and conclusion

In this article, we have studied the comparison of four imputation techniques applied to user-made incomplete data sets with missing covariates. The missingness pattern considered is the monotone pattern. The imputation techniques compared included the last observation carried forward (LOCF) method and three imputation techniques, namely Regression, Propensity Score (PS) and Markov Chain Monte Carlo (MCMC). In order to compare the performance of the methods, we used the originally complete data set (actual data), and then we artificially created missing values to achieve the intended goal. Missingness was imposed on covariates. The results of the regression analysis of the imputed models were compared under three criteria: bias,

efficiency and coverage. Data from a Diabetes study is used to investigate the performance of the considered approaches.

Generally, comparing the analysis based on 20% and 30% missing data rate, among the imputation techniques examined here, the PS and LOCF techniques were notable for consistently producing more biased estimates versus those in the MCMC and Regression methods, regardless of the missing data rate. This agrees with known theoretical findings, that PS can give biased estimates of regression coefficients when data on predictor variables are missing (Allison (2000)). Schafer (1999) found that the PS technique is not appropriate for analyses involving relationships among variables, such as a regression analysis. It would appear that Schafer's (1999) recommendation to not use PS for regression analyses with missing values is supported by the results presented here. On the other hand, in this particular dataset, the LOCF technique gave worse results and less efficient technique, in terms of bias, efficiency and coverage than other imputation techniques considered in the current study. LOCF can be justified from the fact that the method is used primarily for analyses of longitudinal studies that have experienced attrition (Little and Rubin (1987)). Also, Schafer (2000) refers to the LOCF as inferior to regression problems because it ignores regression to the mean. In addition, Molenberghs and Kenward (2007) argued against the method and, noted that the technique makes the strong assumption that there is no change in the subject response between the observed time points and the missing time period, which can lead to biased estimates. However, the LOCF may be appropriate for the particular dataset when the pattern of missingness is monotone.

Findings in general favoured the MCMC and Regression methods over the PS and LOCF methods. As expected, the MCMC and Regression methods yielded better results, regardless of the missingness rate. Both, MCMC and

Regression methods yielded acceptable performance of parameter estimates, and obviously have an advantage over PS and LOCF methods. Furthermore, MCMC and Regression methods yielded estimates closer to each other for bias, efficiency and coverage, and in some cases they yielded the same estimates. This confirmed the theoretical findings from Rubin (1987) and Schafer (1997) who concluded that to handle missing values for a continuous variables in datasets with monotone missing patterns we should use methods that assume multivariate normality as MCMC and Regression.

In conclusion, the results show that we have universally best methods to deal with missing covariates with monotone missingness. From the results, it appears that either MCMC or Regression methods of imputation for estimating regression models with monotone missingness are preferable to PS (which is a non-parametric technique) and LOCF method (which is a single imputation technique).

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